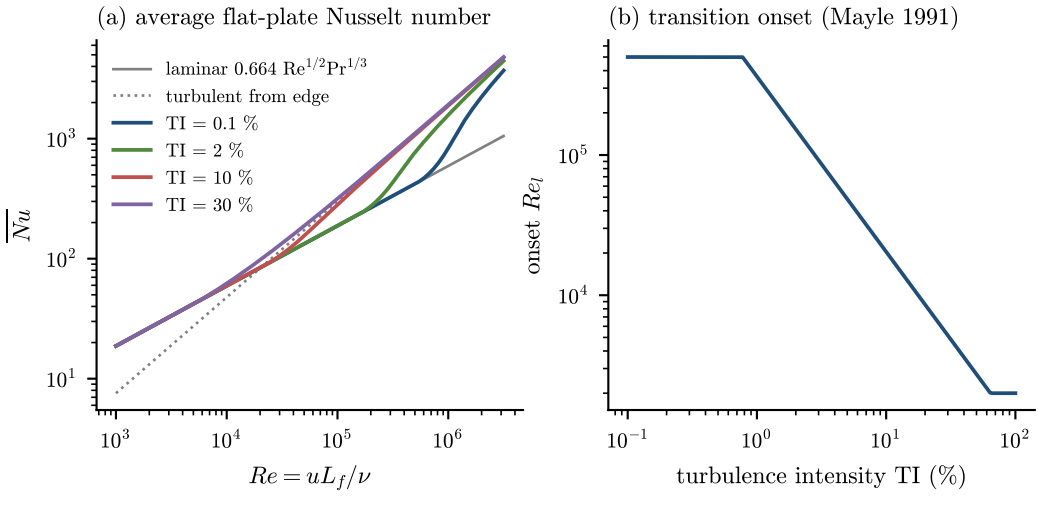
(a) average flat-plate Nusselt number
laminar 0.664 Re^1/2 Pr^1/3
turbulent from edge
TI = 0.1 %
TI = 2 %
TI = 10 %
TI = 30 %
$\overline{Nu}$
10^1
10^2
10^3
10^3
10^4
10^5
10^6
$Re = uL_f/\nu$
(b) transition onset (Mayle 1991)
onset Re_l
10^4
10^5
10^{-1}
10^0
10^1
10^2
turbulence intensity TI (%)